TEXORA · LECTURE SHEET

Production of Weft Knitted Fabrics

From a buyer's specification — fabric type, width, weight, colour — to a running machine: yarn choice by gauge, machine selection, the settings sheet, quality checkpoints during the run, and doffing. This is the production routine that Topics 4–7 assumed and Topic 9 will control.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 8 · LEVEL 3 TERM 1

1. The production sequence

1. **Specification:** structure, finished width and GSM, fibre/yarn (count, twist, lot), colour(s), quantity, delivery form (tube/open width, roll weight).
2. **Machine selection:** structure → bed type (Topics 4–7); width → diameter (open the tube mentally: tube circumference = πD must cover the finished width plus cutting/finishing allowances and expected relaxation change); fineness → gauge matched to yarn count (rule of thumb: the yarn must pass the hook freely with its neighbours — factories use gauge–count charts; e.g. Ne 30 (≈19.7 tex) is the classic E24–E28 jersey yarn).
3. **Yarn preparation:** cones inspected, lots segregated (never mix dye lots in one fabric piece), waxing for coarse/friction-sensitive yarns on some machines, creeling in the planned feeder order (stripes!).
4. **Machine setting:** needles checked, stitch cams set to the calculated stitch length, positive-feed calibrated per feeder, take-down adjusted, counters set for the target piece length/weight.
5. **Trial and approval:** a short length is knitted, relaxed, and measured (width, CPC/WPC, GSM, stitch length by unroving) against the specification; settings corrected before bulk.
6. **Bulk run with patrol inspection:** feeder-wise checks, fault marking, stop-motion function, roll-by-roll records.
7. **Doffing and identification:** roll weighed/length-recorded, labelled with machine, lot, date and operator; grey fabric to inspection/finishing.

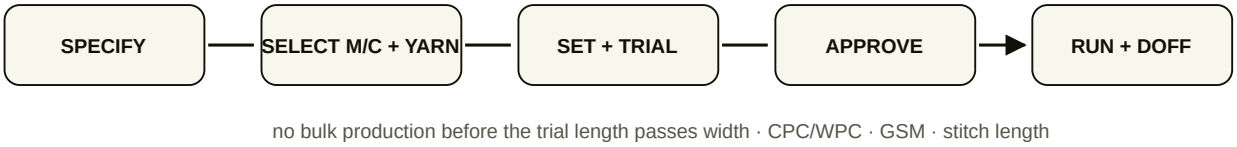


Figure 1 — The production routine. The trial-and-approval gate is where width and GSM errors are still cheap.

2. Width and weight planning — worked example (verified)

WORKED EXAMPLE — WILL THIS MACHINE GIVE THE ORDERED WIDTH AND GSM?

Order: single jersey, open width 175 cm finished, GSM ≈ 150, yarn 19.7 tex.

Machine: D = 26 in, E28 → N = π × 26 × 28 = **2287 needles (wales)**.

Planned relaxed WPC = 13/cm → open width = 2287 ÷ 13 = **175.9 cm** ✓ (tube opened once: all wales lie across the open width).

Set stitch length l = 4.2 mm; expected relaxed CPC = 15/cm. GSM = CPC × WPC × l × tex ÷ 100 = 15 × 13 × 4.2 × 19.7 ÷ 100 = **161.3 g/m²** — slightly heavy. Shorten l to 3.9 mm and CPC rises as loops shrink; re-trial on the machine and lock the settings that actually deliver ≈150 g/m² at 175 cm — the formula plans, the trial decides.

The GSM relation used above is the standard knitted-fabric identity: each loop contributes l (mm) of yarn of linear density tex; loops per m² = CPC × WPC × 10,000; collecting units gives GSM = CPC × WPC × l × tex / 100 with CPC, WPC per cm and l in mm. Topic 12 derives it fully.

3. Stripes and colour work in production

Feeder stripes repeat over the feeder sequence: with 72 feeders, a 6-course colour repeat divides the tube evenly (72 ÷ 6 = 12 repeats per revolution), while a 7-course repeat would drift course by course around the machine — plan repeats that divide the feeder count, or accept the drift where it cannot show (cut panels). Engineered (auto-striper) work changes yarn at a feeder by finger selection; the changeover courses must fall in waste or seam areas of the garment layout. Colour lots: a new dye lot starts at a new roll, never mid-roll.

4. Production records and efficiency

Each roll record carries: machine no., structure, yarn lot(s), feeder plan, stitch length setting, roll weight and length, faults marked, operator and shift. **Machine efficiency** = actual production ÷ theoretical production at the running speed (Topic 12 formulas) — losses come from doffing, yarn breaks, press-off clearing, and waiting. A jersey frame in good order sustains 85–92 %; interlock 80–88 % (Topic 6 example used 85 %). Production meetings review efficiency *with* quality: speed bought with press-offs is not production.

Glossary

Grey (greige) fabric — fabric as knitted, before wet finishing. **Doffing** — removing a completed roll/piece from the machine. **Patrol inspection** — scheduled operator checks during the run. **Settings sheet** — the recorded machine settings for an approved style, reused on repeat orders.

Key points

• Routine: specify → select machine/yarn → set → trial → approve → run → doff, with records at each roll.
• Width is wales ÷ WPC; weight follows GSM = CPC × WPC × l × tex / 100 — planned by formula, locked by trial.
• Stripe repeats must divide the feeder count; dye lots never change mid-roll.

Practice

1. A 30-inch E24 jersey machine (N = 2262) must deliver 160 cm open width. What relaxed WPC is needed?
2. Using the GSM identity, find GSM for CPC 17, WPC 13, l = 4.0 mm, tex 16.4.
3. A stripe repeat of 9 courses is planned on a 72-feeder and on a 60-feeder machine. Which machine keeps the stripe register, and why?
4. List the trial-length measurements that must pass before bulk, and the setting corrected for each if it fails.

Answer key

1. WPC = 2262 ÷ 160 = **14.1 wales/cm**.
2. GSM = 17 × 13 × 4.0 × 16.4 ÷ 100 = **145.0 g/m²**.
3. The 72-feeder machine: 72 ÷ 9 = 8 exact repeats per revolution, so the pattern registers every revolution. On 60 feeders, 60 ÷ 9 is not integral, so the changeover point walks around the tube.
4. Width (→ stitch length/take-down), CPC and WPC after relaxation (→ stitch length), GSM (→ stitch length and yarn), stitch length by unroving one course (→ stitch cams/positive feed), appearance/faults (→ needles, feeders, yarn path). Any complete mapping earns credit.